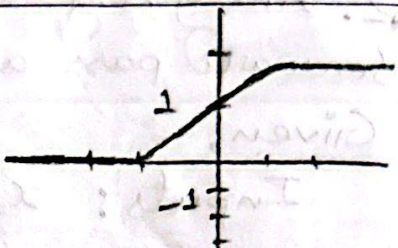
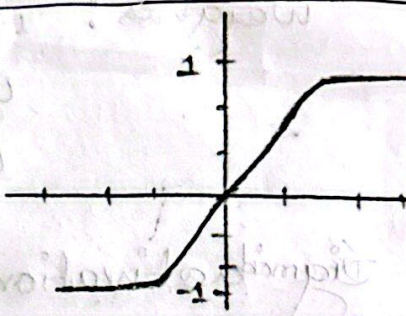
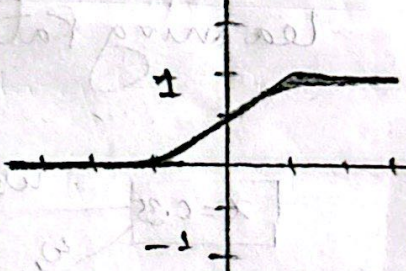
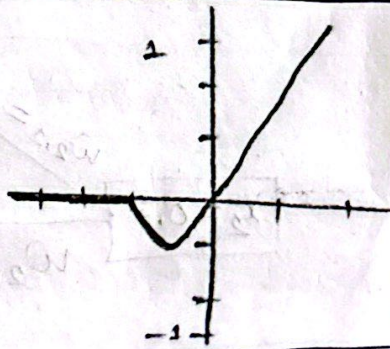


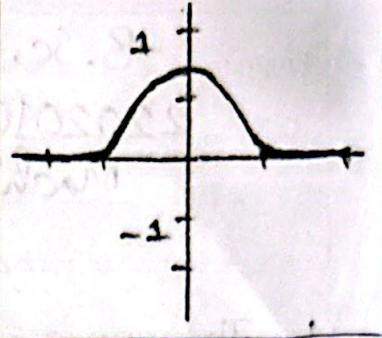


BANGLADESH ARMY UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAUST), SAIDPUR
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Program : B.Sc. in Engg. in CSE Date : 27.11.2025
ID No: 220201065 Level : 4 Term : I Course Code : CSE4139
Course Title: Machine learning

1. The common activation functions are explained below:

Activation Function	Definition	Formula	Figure
Sigmoid/Logistic/softstep	The sigmoid function maps any real input into the range (0,1) which is mostly used for Binary classification	$\sigma(x) = \frac{1}{1 + e^{-x}}$	
Hyperbolic Tangent (tanh)	Scales input between -1 and +1. This function is zero centered and is used in hidden layers.	$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	
Softmax	Converts multiple inputs into a probability distribution, whose sum is 1. It is used in multi-class classification.	$\sigma(x) = \frac{e^x}{\sum_j e^{x_j}}$	
ELISH	It is the Hybrid of ELU and sigmoid that improves gradient flow in deep networks	$f(x) = \begin{cases} \frac{e^x}{1 + e^x} & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$	

Function	Definition	Formula	Figure
Gaussian	Produces bell-shaped output and is used in RBF networks	$f(x) = e^{-x^2}$	

2. Analyzing the network for target output = 0.66 using forward pass and backpropagation.

Given,

Inputs: $x_1 = 0.35$ and $x_2 = 0.7$

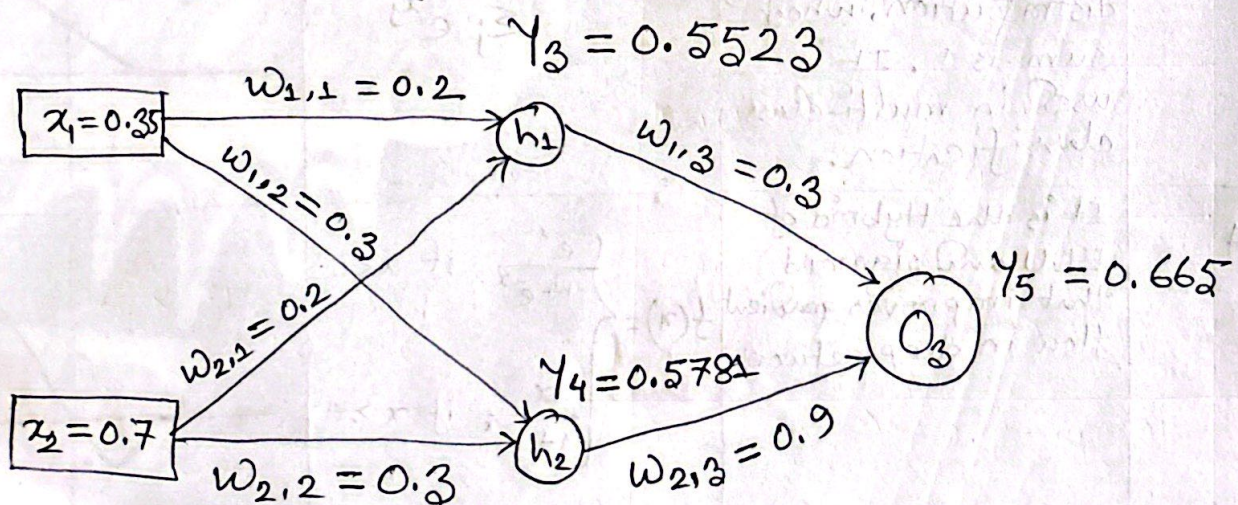
Weights: $w_{1,1} = 0.2$, $w_{2,1} = 0.2$

$w_{1,2} = 0.3$, $w_{2,2} = 0.3$

$w_{1,3} = 0.3$, $w_{2,3} = 0.9$

Sigmoid activation Function: $\sigma(z) = \frac{1}{1 + e^{-z}}$

Learning Rate, $\eta = 1$



Computing the Outputs gives:

$z = \sum (w_{i,j} * x_i)$	$y_j = f(z_j) = \frac{1}{1+e^{-z}}$
$z_1 = (w_{1,1} * x_1) + (w_{2,1} * x_2)$ $= (0.2 * 0.35) + 0.2 * 0.7 = 0.21$	$y_3 = \frac{1}{1+e^{-0.21}} = 0.5523$
$z_2 = (w_{1,2} * x_1) + (w_{2,2} * x_2)$ $= (0.3 * 0.35) + (0.3 * 0.7)$ $= 0.315$	$y_4 = \frac{1}{1+e^{-0.315}} = 0.5781$
$z_3 = (w_{1,3} * y_3) + (w_{2,3} * y_4)$ $= (0.3 * 0.5523) + (0.9 * 0.5781)$ $= 0.686$	$y_5 = \frac{1}{1+e^{-0.686}} = 0.665$

Error Calculation: The actual output is $y_{\text{target}} = 0.66$ but obtained output is $y_5 = 0.665$. Thus, the errors e_j can be calculated as:

$$e_j = y_{\text{target}} - y_5 = 0.66 - 0.665 = -0.005$$

This error can be used for back propagation.

Now, calculating Gradients: The change in each weight is calculated as,

$$\Delta w_{i,j} = \eta * \delta_j * O_i \quad ; \text{ where, } \delta_j \text{ is the error term for each unit and } \eta \text{ is the learning rate.}$$

Output unit errors: For O_j

$$\begin{aligned}\delta_5 &= y_5 * (1 - y_5) (y_{\text{target}} - y_5) \\ &= 0.665 * (1 - 0.665) (0.66 - 0.665) \\ &= -0.00114\end{aligned}$$

The hidden layer unit errors: For h_1 ,

$$\begin{aligned}\delta_3 &= y_3 * (1 - y_3) (\delta_5 * w_{1,3}) = 0.5523 * (1 - 0.5523) (-0.00114 * 0.3) \\ &= -0.085 \times 10^{-3}\end{aligned}$$

For h_2 ,

$$\begin{aligned}\delta_4 &= y_4 * (1 - y_4) (\delta_5 * w_{2,3}) = 0.5781 * (1 - 0.5781) (-0.00114 * 0.3) \\ &= -0.25 \times 10^{-3}\end{aligned}$$

$\therefore$ Updated weights are;

$\begin{aligned}\Delta w_{1,3} &= \delta_5 * y_3 \\ &= (-0.00114 * 0.5523) = -0.00062\end{aligned}$	$\begin{aligned}w_{1,3}(\text{new}) &= 0.3 - 0.00062 \\ &= 0.29938\end{aligned}$	to Output
$\begin{aligned}\Delta w_{2,3} &= \delta_5 * y_4 \\ &= (-0.00114 * 0.5781) = -0.00066\end{aligned}$	$\begin{aligned}w_{2,3}(\text{new}) &= 0.3 - 0.00066 \\ &= 0.29934\end{aligned}$	
$\begin{aligned}\Delta w_{1,1} &= \delta_3 * x_1 \\ &= -8.5 \times 10^{-5} * 0.35 = 0.00003\end{aligned}$	$\begin{aligned}w_{1,1}(\text{new}) &= 0.2 - 0.00003 \\ &= 0.19997\end{aligned}$	to hidden layer
$\begin{aligned}\Delta w_{2,1} &= \delta_3 * x_2 \\ &= -8.5 \times 10^{-5} * 0.7 = -0.00006\end{aligned}$	$\begin{aligned}w_{2,1}(\text{new}) &= 0.2 - 0.00006 \\ &= 0.19994\end{aligned}$	
$\begin{aligned}\Delta w_{1,2} &= \delta_4 * x_1 \\ &= -2.5 \times 10^{-4} * 0.35 = -0.00009\end{aligned}$	$\begin{aligned}w_{1,2}(\text{new}) &= 0.3 - 0.00009 \\ &= 0.29991\end{aligned}$	
$\begin{aligned}\Delta w_{2,2} &= \delta_4 * x_2 \\ &= -2.5 \times 10^{-4} * 0.7 = -0.00018\end{aligned}$	$\begin{aligned}w_{2,2}(\text{new}) &= 0.3 - 0.00018 \\ &= 0.29982\end{aligned}$	